

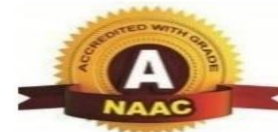


S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR

(An Autonomous Institute, Affiliated to R.T.M. Nagpur University)

DEPARTMENT OF EMERGING TECHNOLOGIES CSE (AI&ML)

"Become an excellent center for Emerging Technologies in Computer Science to create competent professionals"



COURSE NAME: Python Programming Lab

SESSION: 2026-27(ODD)

COURSE CODE: N-PCCCM304P

SEMESTER: III

NAME OF STUDENT: _____ ROLL NUMBER: _____

INDEX

PR. No	Aim of Practical	Date of Performance	Date of Submission	Performance (5M)	Journal Submission & Viva (3M)	Attendance (2M)	Total (10M)	Signature
Pre Lab	An Introduction to Python Programming Lab							
1	Construct a program to generate marksheet of a student's containing following details a) Name, b) Department, c) Year d) Semester e) Section f) Roll Number g) Marks obtained in 5 subjects in particular semester h) Total marks i) Percentage j) Grade							
2	Construct a program to read the temperatures of water in degree Celsius from the user and print the appropriate output depending upon the following conditions: a) If the temperature is less than -273.15°C, print that the temperature is invalid because it is below absolute zero. b) If it is exactly -273.15°C, print that the temperature is "absolute 0". c) If the temperature is between -273.15°C and 0°C, print that the temperature is below "freezing". d) If it is 0°C, print that the temperature is at the "freezing point". e) If it is between 0°C and 99°C, print that the temperature is in the "normal range". f) If it is equals to 100°C, print that the temperature is at the "boiling point". g) If it is above 100°C, print that the temperature is above the "boiling point".							

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3	a) You are given three different integer numbers. Write a program which will find the largest of these three numbers. b) Construct a program which will print the statement “Hello Students” twenty times by using i) while loop ii) for loop c) Calculate the sum of all numbers from 1 to n given number. 'n' will be given by user.							
4	Sample Python dictionary data and list labels: <pre>exam_data = {'name': ['Ajay', 'Vijay', 'Vivek', 'Atharv', 'Apurva', 'Rupal', 'Priyal', 'Shrushti', 'Vishal', 'Pranay'], 'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19], 'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1], 'qualify': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}</pre> labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'] a) Construct a program to select the 'name' and 'score' columns from the above DataFrame. b) Construct a program to select the rows in which the score is between 15 to 20 from the above DataFrame. c) Create a program to select the rows where the score is missing in the above DataFrame. d) Construct a program to change the score in row 'd' to '11.5' in the above DataFrame e) Develop a program to calculate the sum of the examination attempts by the students.							
5	Develop a program that asks the user to enter a series of integers and store them in a Tuple. Perform the following: a) Print the total number of items in the Tuple. b) Print the last item in the Tuple. c) Print the Tuple elements in reverse order. d) Print Yes if the Tuple contains an integer '5' and 'No' otherwise. e) Remove the first and last items from the Tuple, sort the remaining items, and print the result.							

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6	Create functions for the following: a) Function that takes two numbers as input and returns their sum. b) Function to calculate the area of a circle. c) Function that reverses a given string. d) Function that takes an integer and returns its square and square root. Explore the use of default arguments and keyword arguments in functions.							
7	Create a Menu Driven program for showing details of a Bank Account by implementing different functions for the following: a) Display the current Balance b) Display the statement of last 1 month. c) Display the statement of last 3 months. d) Display the statement of Current Financial Year							
8	"You are given a space separated list of nine integers. Your task is to convert this list into a 3X3 NumPy array. Input Format 1 2 3 4 5 6 7 8 9 Output Format Print the 3X3 NumPy array.							
9	Create a Panda Dataframe script by reading a file "books.csv".The "books.csv" contains information regarding the books such as title, name of author, edition, Publication year and publishing house, price. Create an application to perform the following operations: a) Print the complete report of books in a Tabular form. b) Print the list of available books of a given author c) Print the list of available books of a given publishing house d) Print the Titles of cheapest & costliest book available e) Print the list by sorting based on the year of publication and cost of the book							

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10	Perform the following File Handling Operations: a) Construct a program to read a text file and print its contents. b) Create a program that writes a list of names to a file. c) Develop a program to read a file, count the occurrences of each word, and write the word counts to a new file. d) Experiment with different file modes ('r', 'w', 'a', 'r+').							
11	Create a program that reads "employee.xlsx" file of Infosys Software Solutions which includes columns such as Employee ID, Employee Name, Department, Designation etc. Construct a program to print the following reports: a) Print the list of employees working for "Automotive" domain. b) Print the details of an employee with employee ID given by an end user. d) Print the list of all the Developers of Infosys.							
Post Lab	Open Ended Practical using AI tools.							

Signature of Lab Batch Course In-charge